SEAT No.________ SARDAR PATEL UNIVERSITY No. of Printed Pages : 02

M. Sc.(IT Integrated) EXAMINATION, IV SEM

Date : 17th Sep 2020, Thursday | Time :03:00pm To 05:00pm
Session : Evening | Sub: Computer based numerical & statistical
Course No : PS04FIIT01 | Total marks : 70

Q – 1 Multiple Choice Question [10]

i) If for a real continuous function $f(x)$, $f(a)f(b)<0$, then in the range of [a,b] for $f(x)=0$, there is (are).
 a) One root b) an undeterminable number of roots.
 c) No roots d) At least one root

ii) $f(a) < 0$, $f(b) > 0$ and if $x_0 \in (a,b)$ is first approximation with $f(x_0) < 0$ then in bisection method,
 a) x_0 is to be replaced by a b) a is to be replaced by c
 c) b is to be replaced by x_0 d) x_0 is to be replaced by b

iii) The number 0.01850×10^3 has ________ significant digits Referential
 a) 3 b) 4 c) 5 d) 6

iv) To estimate the value of dependent variable x for a given value of independent variable y, the process is known as _______.
 a) Extrapolation b) Inverse Interpolation
 c) Interpolation d) Polynomial

v) ________ is called the backward difference operator.
 a) Δ b) ∇ c) $\emptyset$ d) $\cup$

vi) y depends on x can be written as _________
 a) f(x) or y_x b) f(xy) c) f(yx) d) None

vii) We can find solution of system of linear, algebraic equations using ________
 a) Newton-Raphson method b) Bisection method
 c) Gauss-Seidel method d) None

viii) The system of linear equation AX = B can be solved by matrix inversion method only if______
 a) $A \neq 0$ b) $|A| \neq 0$ c) $|A| = 0$ d) A is symmetric

ix) Forecasts _______
 a) become more accurate with longer time horizons b) are rarely perfect
 c) are more accurate for individual items than for groups of items d) all of the above

x) Gradual, long-term movement in time-series data is called _________
 a) seasonal variation b) cycles
 c) trends d) exponential variation

Q – 2 Fill in the Blanks and True – False [08]

i) The lagranges method is not iterative method. (True / False)
ii) For real root of an equation x3-2x-5=0 the root lies between________.
iii) Theile defines Extrapolation as "The art of reading between the lines of a table." (True / False)
iv) All the formulae of interpolation are based on the fundamental assumption that the given data can be expressed as a ____________.
v) Rate of change of distance with respect to time represents Acceleration. (True / False)
vi) The system of linear equation AX=B is said to be non-homogeneous if ________
vii) A ________ of a time series attached to long-term variations is term as Secular trend
viii) A time series consist of four components. (True / False)

(1) (P.T.O.)

Q – 3 Short Answer **attempt any ten (Each carry 2 marks)** [20]

i) Find the next iterative value of the root of using secant method, if the initial guesses are 3 and 4

ii) Use the False Position method to obtain initial solution of $x^3 - 9x + 1 = 0$

iii) Describe the stopping rules to obtain approximate solution for given non-linear equations.

iv) If x lies in the upper half of the table and if $x = x_k$, then what is $\frac{dy(x)}{dx}$ and $\frac{d^2y(x)}{dx^2}$?

v) Define Interpolation.

vi) Write algorithm for Backward difference table

vii) List the component of Time series.

viii) Solve the system of equation: $\begin{matrix} 2x + 3y = -10 \\ -x + 4y = -4 \end{matrix}$

ix) If x lies in the upper half of the table and if $x = x_k$, then what is $\frac{dy(x)}{dx}$ and $\frac{d^2y(x)}{dx^2}$?

x) What is Time Series?

xi) What do you mean by Secular trend?

xii) What do you mean by random or irregular Variation?

Q – 4 Attempt any four **(Each carry 8 marks)** [32]

i) Write algorithm for bisection and regular Falsi method.

ii) Write algorithm for iterative method. And define relative error.

iii) Estimate by Newton's method of interpolation, the expectation of life at age 32 from the following table.

Age (In Years)	10	15	20	25	30	35
Expectation of life (In Year)	35.3	32.4	29.2	26.1	23.2	20.5

iv) Given the table of values as (use divided difference method) Find y(2.35)

x	2.5	3.0	3.5	4.75	6.0
y(x)	8.85	11.45	20.66	22.85	38.60

v) Explain the Gauss – Sedel method for solution of system of linear equation.

vi) The distance (s) covered by a car in given time (t) is given in the following table:

Time(minutes)	10	12	14	16	18
Distance(km)	12	15	20	27	37

Determine the speed of the car at t=13 minutes.

vii) Obtain seasonal indices using simple average method.

Year	Q-1	Q-II	Q-III	Q-IV
1990	30	81	62	119
1991	33	104	86	171
1992	42	153	99	221
1993	56	172	129	235

viii) Obtain seasonal indices using ratio to moving average.

Year	Q-1	Q-II	Q-III	Q-IV
1990	30	40	36	34
1991	34	52	50	44
1992	40	58	54	48
1993	54	76	68	62

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